

As questões de 1 a 6 referem-se ao texto a seguir:

THE MAN IN THE CONVERTIBLE

- 1 One morning, well after I was diagnosed with cancer, I got an email from Robbee Kosak, Carnegie Mellon's vice president for advancement. She told me a story.
- She said she had been driving home from work the night before, and she found herself behind a man in a convertible. It was a warm, gorgeous, early-spring evening, and the man had his top down and all his windows lowered. His arm was hanging over the driver's side door, and his fingers were tapping along to the music on his radio. His head was bobbing along, too, as the wind blew through his hair.
- 5 Robbee changed lanes and pulled a little closer. From the side, she could see that the man had a slight smile on his face, the kind of absentminded smile a person might have when he's all alone, happy in his own thoughts. Robbee found herself thinking: "Wow, this is the epitome of a person appreciating this day and this moment."
- 10 The convertible eventually turned the corner, and that's when Robbee got a look at the man's full face. "Oh my God," she said to herself. "It's Randy Pausch!"
- She was so struck by the sight of me. She knew that my cancer diagnosis was grim. And yet, as she wrote in her email, she was moved by how contented I seemed. In this private moment, I was obviously in high spirits. Robbee wrote in her email: "You can never know how much that glimpse of you made my day, reminding me of what life is all about."
- 15 I read Robbee's email several times. I came to look at it as a feedback loop of sorts.
- It has not always been easy to stay positive through my cancer treatment. When you have a dire medical issue, it's tough to know how you're really faring emotionally. I had wondered whether a part of me was acting when I was with other people. Maybe at times I forced myself to appear strong and upbeat. Many cancer patients feel obliged to put up a brave front. Was I doing that, too?
- 20 But Robbee had come upon me in an unguarded moment. I'd like to think she saw me as I am. She certainly saw me as I was that evening.
- Her mail was just a paragraph, but it meant a great deal to me. She had given me a window into myself. I was still fully engaged. I still knew life was good. I was doing OK.
- 25

Fonte: PAUSCH, R. The last lecture. New York, Hyperion, 2008. p.64-65.

Questão 1. O autor do texto

- A () utiliza a dissertação e a descrição como tipologia textual predominante.
- B () é narrador observador e mescla discursos direto, indireto e indireto livre.
- C () usa foco narrativo em primeira pessoa configurando o texto autobiográfico.
- D () utiliza linguagem coloquial nos diálogos para externar seus sentimentos.
- E () recorre a figuras de pensamento para compor o gênero dramático.

Questão 2. De acordo com as informações no texto, Robbee Kosak

- A () descreveu detalhadamente o cenário do seu primeiro encontro com Randy Pausch.
- B () sentiu-se atraída pelo veículo de Randy Pausch devido à alta velocidade dele.
- C () escreveu palavras motivadoras a Randy Pausch porque desejava reanimá-lo.
- D () caracterizou o motorista do veículo como uma pessoa satisfeita e de bem com a vida.
- E () ocupava o cargo de Vice-Presidente na empresa presidida por Randy Pausch.

Questão 3. Assinale a oração que **não** contém expressão ou termo qualificador.

- A () It was a warm, gorgeous, early-spring evening... (linha 4)
- B () She was so struck by the sight of me. (linha 13)
- C () ...she was moved by how contented I seemed. (linha 14)
- D () ...I was obviously in high spirits. (linha 14 e 15)
- E () She certainly saw me as I was that evening. (linhas 22 e 23)

Questão 4. A frase “**She had given me a window into myself**” (linha 24 e 25) expressa

- A () percepção que Robbee Kosak transmitiu de si própria para Pausch.
- B () visão reduzida que o autor transmitiu sobre seu lado otimista.
- C () aparência distorcida de uma personalidade extrovertida.
- D () constatação de que Randy Pausch não transmitia vontade de viver.
- E () percepção do narrador sobre algo de que ele não se dava conta.

Questão 5. Com relação às escolhas lexicais do autor no texto, pode-se afirmar que

- A () em “Carnegie Mellon’s vice president” (linhas 1 e 2), “the driver’s side door” (linha 5), “the man’s full face” (linha 11) e em “Robbee’s email” (linha 17) o uso do ’s indica caso possessivo.
- B () em “she found herself behind a man in a convertible” (linhas 3 e 4) e em “Robbee found herself thinking...” (linha 9), o verbo *find* pode ser substituído por *realize* sem prejudicar o sentido.
- C () em “as the wind blew through his hair” (linha 6), “...as she wrote in her email...” (linhas 13 e 14) e em “...as a feedback loop of sorts” (linha 17) os itens sublinhados podem ser substituídos por *while* sem prejudicar o sentido.
- D () *absentminded* (linha 8), *feedback* (linha 17) e *engaged* (linha 25) são empregados como substantivos.
- E () *eventually* (linha 11), *obviously* (linha 14), *really* (linha 19) e *certainly* (linha 23) indicam o mesmo tipo de advérbio.

Questão 6. Na frase “**She said she had been driving home from work the night before, and she found herself behind a man in a convertible**” (linhas 3 e 4), a formação correta quanto ao uso do discurso direto é:

- A () She said: “I was driving home from work last night, and I found myself behind a man in a convertible”.
- B () She said: “I had been driving home from work last night, and I found me behind a man in a convertible”.
- C () She said: “I drove home yesterday night from work, and I had found myself behind a man in a convertible”.
- D () She said: “I had driven home the night before, and I found myself behind a man in a convertible”.
- E () She said: “I was driving home from work yesterday, and I was finding myself behind a man in a convertible”.

As questões de 7 a 10 referem-se ao texto a seguir:

IRON MAN DESIGNERS TO BUILD BODY ARMOUR FOR US ARMY

Hollywood special effect team is working on a new Iron Man 'agile exoskeleton' for US soldiers

1 The Oscar-nominated special effects team behind the Iron Man suit has been contracted to design body armour for the US military.

Legacy Effects, a Hollywood design studio based in California, has previously worked on power suits for films such as RoboCop, Captain America, The Terminator and Iron Man. Now, the company is building
5 body armour equipped with an "agile exoskeleton" that will allow soldiers to carry hundreds of pounds of equipment, the Wall Street Journal reports.

"We are trying to be revolutionary," said Mike Fieldson, who manages the US military project known as the Tactical Assault Light Operator Suit (Talos).

10 Three prototypes have been presented to the Pentagon by teams of bioengineers, technologists and a Canadian company that studies insect and animal exoskeletons. The prototypes will contribute to the creation of a new generation of body armour which the US Special Operations Command aims to complete within four years.

The suits are designed to protect soldiers from bullets, explosions and bayonet attacks.

15 Legacy Effects admits that bringing an Iron Man to life presents significant challenges. For one thing, a real-life version of the suit would add extra bulk to a soldier limiting his or her agility. Also, the company estimates that the Iron Man suit would probably weigh about 180kg, and would need to be supported by a mobile exoskeleton, but "none of the exoskeletons in the industry are capable of moving that much weight", SlashGear reports.

20 Russ Angold of Ekso Bionics, a company that designs exoskeletons for medical use, says that power armour in films offer an unrealistic model, so engineers are presently trying to make the suits more practical. "Hollywood has definitely made the Iron Man suit impossibly thin, impossibly light, impossibly agile and impossibly energy efficient. So we're really trying to solve the problem and ask the question: What would Iron Man look like if it was real?"

25 The US military has so far spent about \$10 million on Talos, prompting the armed services committee to request a briefing on the project to ensure taxpayer money is not being wasted.

"Will you ever have an Iron Man? I don't know," said Brian Dowling, a former soldier involved in the project. "But you'll have some greatly improved technology along the way".

Fonte: <http://www.theweek.co.uk/world-news/59323/iron-man-designers-to-build-body-armour-for-us-army>

Acesso: 13/ago/2014

Questão 7. O projeto Talos

- A () tem por objetivo construir uma prótese a ser usada por soldados americanos.
- B () foi idealizado há quatro anos e três protótipos foram apresentados.
- C () é constituído por uma equipe formada por militares americanos e pesquisadores aposentados.
- D () conta com a participação do *studio* que desenvolveu a armadura do Iron Man.
- E () faz parte de um projeto mais amplo desenvolvido pela empresa americana Legacy Effects.

Questão 8. A empresa Legacy Effects

- A () tem experiência em criar roupas especiais para filmes americanos famosos.
- B () tem como sede o estado da Califórnia e prevê um gasto de 10 milhões no projeto Talos.
- C () é uma das parceiras do exército americano na idealização de exoesqueleto para uso médico.
- D () aceitou o desafio do projeto Talos e garante cumprir todos os objetivos que o projeto impõe.
- E () baseou-se nos estudos de exoesqueletos de animais e de insetos para criar o protótipo americano.

Questão 9. A expressão sublinhada no trecho "...but “none of the exoskeletons in the industry are capable of moving that much weight”” (linha 17) não pode ser substituída por

- A () are able to move.
- B () are keen on moving.
- C () have the ability to move.
- D () are succeed in moving.
- E () can move.

Questão 10. A vestimenta idealizada no projeto Talos deverá satisfazer apenas uma das condições abaixo:

- A () não ultrapassar o orçamento de 10 milhões de dólares previsto pelo governo americano.
- B () ajustar-se ao corpo humano independentemente do peso e do tamanho do usuário.
- C () oferecer condições de realizar operações militares carregando muito peso.
- D () auxiliar o soldado em combate, aumentando o tempo em incursões militares não motorizadas.
- E () ser funcional e conter bateria duradoura e recarregável por energia solar.

As questões de 11 a 17 referem-se ao texto a seguir:

STICKERNOMICS

Football albums

Got, got, got, got, got, need

1 THE World Cup is still two weeks away, but for children worldwide (plus disturbing numbers of adults) the race to complete the Brazil 2014 sticker book started long ago. Panini, an Italian firm, has produced sticker albums for World Cups since Mexico 1970; this year's version has 640 stickers to collect. Collecting them is no idle pursuit, however. Getting every slot filled delivers an early lesson in probability, the value of
5 statistical tests and the importance of liquidity.

When you start an album, your first sticker (in Britain, they come in packs of five) has a 640/640 probability of being needed. As the spaces get filled, the odds of opening a pack and finding a sticker you want fall. According to Sylvain Sardy and Yvan Velenik, two mathematicians at the University of Geneva, the number of sticker packs that you would have to buy on average to fill the album by mechanically buying pack
10 after pack would be 899. That assumes there is no supply shock to the market (the theft of hundreds of thousands of stickers in Brazil in April left many fearful that Panini would run short of cards).

It also assumes that the market is not being rigged. Panini says that each sticker is printed in the same volumes and randomly distributed. In a 2010 paper Messrs Sardy and Velenik gamely played the role of "regulator" by checking the distribution of stickers for a 660-sticker album sold in Switzerland for that year's
15 World Cup. Out of their sample of 6,000 stickers, they expected to see each sticker 9.09 times on average (6,000/660), which was broadly borne out in practice.

Even in a fair market, it is inefficient to buy endless packs as an individual (not to mention bloody expensive for the parents). The answer is to create a market for collectors to swap their unwanted stickers. The playground is one version of this market, where a child who has a card prized by many suddenly
20 understands the power of limited supply. Sticker fairs are another. As with any market, liquidity counts. The more people who can be attracted into the market with their duplicate cards, the better the chances of finding the sticker you want.

Messrs Sardy and Velenik reckon that a group of ten astute sticker-swappers would need a mere 1,435 packs between them to complete all ten albums, if they take advantage of Panini's practice of selling the final 50 missing stickers to order. Internet forums, where potentially unlimited numbers of people can swap stickers, make this number fall even further. The idea of a totally efficient market should dismay Panini,
25 which will sell fewer packs as a result. But as in all markets, behaviour is not strictly rational. Despite entreaties, your correspondent's son is prepared to tear out most of his stickers to get hold of Lionel Messi.

Fonte: <http://www.economist.com/news/finance-and-economics/21603019-got-got-got-got-got-need-stickernomics>

Acesso: 13/ago/2014

Questão 11. O autor do texto

- I. atribui ao roubo de milhares de figurinhas no Brasil a dificuldade para compra e troca entre colecionadores.
- II. deprecia as estratégias do Grupo Panini para comercializar álbuns de figurinhas da Copa do Mundo.
- III. descreve o mercado de figurinhas da Copa do Mundo e apresenta aos colecionadores possibilidades de obtenção de figurinhas.

Está(ão) correta(s)

- A () apenas a I.
- B () apenas a II.
- C () apenas a III.
- D () apenas I e II.
- E () apenas I e III.

Questão 12. De acordo com o texto,

- A () a empresa Panini comercializa álbuns de figurinhas da Copa do Mundo há 30 anos.
- B () é impossível completar o álbum sem que os colecionadores recorram a feiras e redes sociais.
- C () são necessárias 1.500 figurinhas para completar um álbum.
- D () a empresa Panini disponibiliza a venda das 50 figurinhas faltantes aos colecionadores.
- E () o processo de confecção e distribuição das figurinhas é feito aleatoriamente pela Panini.

Questão 13. Assinale a opção em que a construção verbal está na voz ativa.

- A () Panini [...] has produced sticker albums for... (linhas 2 e 3)
- B () As the spaces get filled... (linha 7)
- C () ...each sticker is printed in the same... (linha 12)
- D () ...which was broadly borne out in practice. (linha 16)
- E () ...where a child who has a card prized by many... (linha 19)

Questão 14. Marque a opção em que o uso do **ing** denota ação contínua.

- A () ...disturbing number of adults... (linha 1)
- B () Collecting them is no idle pursuit... (linhas 3 e 4)
- C () ...your first sticker [...] has a 640/640 probability of being needed. (linhas 6 e 7)
- D () According to Sylvain Sardy and Yvan Velenik... (linha 8)
- E () ...the market is not being rigged. (linha 12)

Questão 15. Marque a opção em que o item lexical sublinhado **não** remete a uma informação anterior.

- A () That assumes there is no supply... (linha 10)
- B () ...left many fearful that Panini would run short of cards... (linha 11)
- C () ...sold in Switzerland for that year's World Cup. (linhas 14 e 15)
- D () ...one version of this market, where a child who has a card prized... (linha 19)
- E () ...should dismay Panini, which will sell fewer packs... (linhas 26 e 27)

Questão 16. De acordo com o texto, Sardy e Velenik

- A () insistem a formação de grupos de 10 colecionadores para facilitar o preenchimento total de álbuns de figurinhas.
- B () fiscalizam a compatibilidade entre a produção de figurinhas e sua comercialização desde 2010.
- C () verificaram na Suíça a repetição de aproximadamente 9 vezes cada figurinha em um lote de 6.000 figurinhas.
- D () são matemáticos pesquisadores da empresa Panini, responsáveis pela distribuição das figurinhas.
- E () consideram que as práticas de obtenção de figurinhas da Copa do Mundo são injustas e manipuláveis.

Questão 17. Em “**Despite entreaties, your correspondent’s son is prepared to tear out most of his stickers to get hold of Lionel Messi**” (linhas 27 e 28), depreende-se que o autor

- A () reconhece que também faz parte do grupo de colecionadores fanáticos por álbuns de figurinhas.
- B () se dispõe a tudo para conseguir a figurinha de Lionel Messi para o filho.
- C () busca adquirir as figurinhas mais disputadas para seu filho por meio das redes sociais.
- D () inclui-se no grupo de colecionadores insensatos de figurinhas de Copa de Mundo 2014.
- E () vivencia em casa o esforço de um colecionador para obter uma única figurinha.

As questões de 18 a 20 referem-se à tirinha a seguir:



Fonte: <http://www.pleated-jeans.com/2011/12/27/the-60-funniest-web-comics-of-2011/> Acesso em 25 de julho de 2014.

Questão 18. Marque a opção que pode substituir “**due to**” sem alterar o sentido do período.

- A () by means of B () in case of C () in spite of
D () instead of E () because of

Questão 19. A relação semântica entre os dois quadros é de

- A () *corroboration.* B () *contradiction.* C () *substantiation.*
D () *establishment.* E () *reinforcement.*

Questão 20. A reportagem anunciada na tirinha

- A () mostra a relação direta entre obesidade e consumo de produtos calóricos.
B () divulga um estudo científico com o objetivo de mudar o comportamento da audiência.
C () demonstra indiferença com relação ao tema.
D () estimula o uso de redes sociais para divulgar produtos dos patrocinadores.
E () revela que 70% das pessoas com sobrepeso são sedentárias.

FIM DA PROVA

**PARE AQUI SE AINDA NÃO QUISE
VER AS RESPOSTAS.**

**O GABARITO COMEÇA NA PRÓXIMA
PÁGINA.**



INSTITUTO TECNOLÓGICO DE AERONÁUTICA

VESTIBULAR 2015

GABARITO

Física		Inglês		Português		Matemática		Química	
1	C	1	C	21	D	1	D	1	A
2	D	2	D	22	E	2	C	2	E
3	C	3	E	23	D	3	D	3	C
4	C	4	E	24	A	4	E	4	D
5	C	5	A	25	D	5	D	5	E
6	B	6	A	26	C	6	E	6	B
7	B	7	D	27	B	7	B	7	E
8	A	8	A	28	A	8	B	8	B
9	C	9	*	29	A	9	C	9	D
10	E	10	C	30	B	10	A	10	C
11	A	11	C	31	C	11	B	11	E
12	B	12	D	32	D	12	E	12	D
13	B	13	A	33	C	13	C	13	D
14	A	14	E	34	B	14	D	14	C
15	D	15	B	35	E	15	E	15	D
16	A	16	C	36	C	16	A	16	B
17	C	17	E	37	E	17	E	17	B
18	C	18	E	38	A	18	A	18	B
19	E	19	B	39	E	19	A	19	C
20	D	20	C	40	A	20	C	20	C

Observação: a opção D da questão 9 da prova de inglês apresenta erro de digitação, o que a torna gramaticalmente incorreta e pode confundir o candidato. Desta forma a questão foi anulada e considerada correta para todos os candidatos.